

Strong and weak acids and bases

Learning outcomes

By the end of this section you should be able to recognise that acid–base equilibria lie in the direction of the weaker conjugate.

If you measure the pH of 0.1 mol dm^{-3} solutions of lemon acid and battery acid you would obtain different values. The pH of lemon juice would be around 2–3 while that of battery acid would be around 0–1. Why? What is the difference between the two acids?

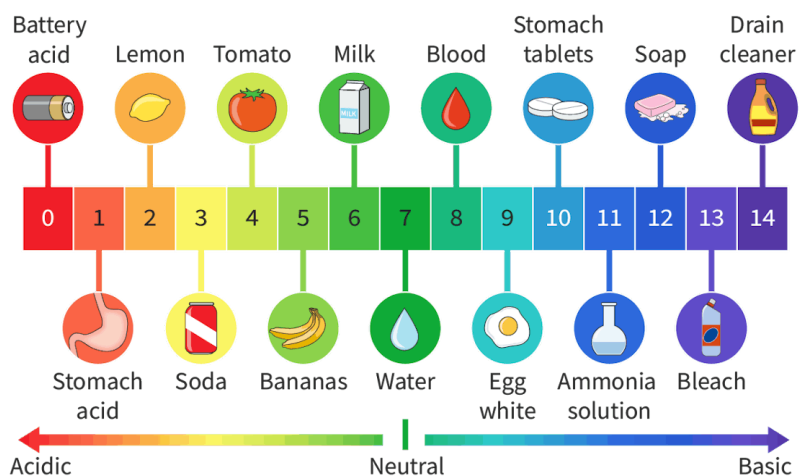


Figure 1. The pH of battery acid and lemon juice are different.

More information for figure 1

The image shows a pH scale ranging from 0 to 14, marking the acidic, neutral, and basic regions. At pH 0, battery acid is depicted with an image of a battery. Lemon juice, with a lemon icon, corresponds to pH 2. Other substances shown include tomato at pH 4, milk at pH 6, and blood at pH 7, the neutral point. The diagram continues with substances like stomach tablets at pH 9, soap at pH 12, and drain cleaner at pH 14. An arrow at the bottom indicates the progression from acidic to basic.

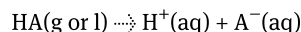
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Acids and bases can be classified as being strong or weak. The strength of an acid refers to its degree of dissociation (or ionisation) in aqueous solution. Strong acids and bases completely dissociate (or ionise) in solution whereas weak acids and bases only partially dissociate (or ionise) in solution.

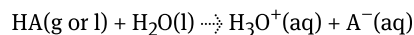
Strong and weak acids

We saw in section R3.1.4-5a ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)) that when a strong acid dissolves in water, it completely dissociates into ions; that is, all the molecules of acid react with the water to produce hydronium ions (H_3O^+), or

hydrogen ions (H^+). The general equation for the dissociation of a strong acid, HA, is given below. Note that a one-way arrow ($\rightarrow$) is used to show the complete dissociation (or ionisation) of the acid.



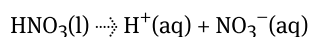
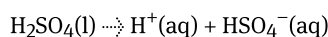
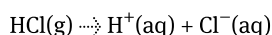
This reaction can also be written with the hydronium ion (H_3O^+) included:



Study skills

For a strong acid, the concentration of H^+ ions is equal to the initial concentration of the strong acid.

The equations for the dissociation of the three common strong acids, hydrochloric, sulfuric and nitric acids, are shown below:



Note that sulfuric acid, H_2SO_4 , is diprotic (it contains two acidic hydrogens). However, only the first dissociation can be considered strong.

Study skills

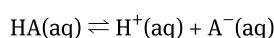
HCl, HBr, HI, HNO_3 and H_2SO_4 are strong acids.

Organic acids, such as ethanoic acid, are usually weak acids. When ethanoic acid is added to water, only a relatively small percentage (1% or less) of its molecules dissociate to form H_3O^+ ions. An equilibrium is established in which the majority of the acid molecules do not dissociate into ions. In other words, the equilibrium position lies very much to the left and the value of the equilibrium constant, K_c , for the dissociation is small.

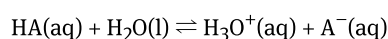
Study skills

For a weak acid (or weak base) the equilibrium position lies to the left and the value of the equilibrium constant, K_c , for the dissociation is small.

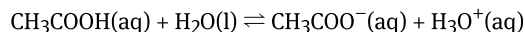
The general equation for the dissociation of a weak acid HA is shown below. Note the use of the equilibrium sign ($\rightleftharpoons$) in the equation.



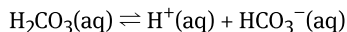
This reaction can also be written involving the formation of the hydronium ion (H_3O^+):



Common examples of weak acids are ethanoic acid (CH_3COOH) and carbonic acid (H_2CO_3), which dissociate according to the following equations:

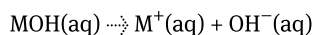


or

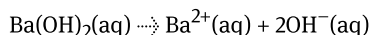
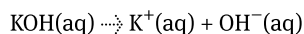
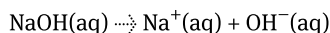


Strong and weak bases

A strong base (or alkali) is one which completely dissociates (or ionises) into ions in aqueous solution. Strong bases include the metal hydroxides of group 1 (LiOH and NaOH) and barium hydroxide ($\text{Ba}(\text{OH})_2$). The general equation for the dissociation of a strong base, MOH , is shown below.



The three common strong bases are sodium hydroxide, potassium hydroxide and barium hydroxide, which dissociate according to the following equations:



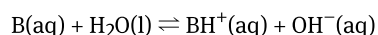
Like strong acids, the equilibrium position for the dissociation of a strong base lies very much to the right. For this reason, a 1 mol dm^{-3} solution of NaOH will contain 1 mol dm^{-3} of OH^- ions. Note that the dissociation of one mole of barium hydroxide, $\text{Ba}(\text{OH})_2$, produces two moles of OH^- ions.

Study skills

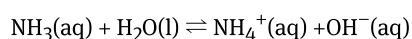
Group 1 hydroxides are strong bases.

For a strong base, the concentration of OH^- ions, $[\text{OH}^-]$, is equal to the initial concentration of the strong base.

All other bases, except for the hydroxides of groups 1 and 2, are weak. Weak bases are made up of molecules that partially dissociate in water to release hydroxide ions. The general equation for the dissociation of a weak base, B , is shown below.



At equilibrium, the majority of the weak base molecules have not dissociated into ions. In other words, the equilibrium position lies very much to the left and the value of the equilibrium constant, K_c , for the dissociation is small. Examples of weak bases include aqueous ammonia (NH_3) and amines such as ethanamine ($\text{C}_2\text{H}_5\text{NH}_2$). The equations for the dissociation of these weak bases are shown below.



Study skills

When defining a strong or weak acid or base, use the terms complete and partial to describe their dissociation (or ionisation). Also, remember to use an equilibrium sign ($\rightleftharpoons$) for weak acids and bases.

For weak acids and bases, the equilibrium concentrations of H_3O^+ ions, or OH^- ions, are much lower than for strong acids and bases. These differences in concentration can be used to distinguish between solutions of weak and strong acids and bases.

The calculations involved in determining the pH of a strong acid were covered in section R3.1.4–5a ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)). Recall that when calculating the pH of a strong acid or base, the $[\text{H}^+]$ or $[\text{OH}^-]$ are equal to the initial concentration of the strong acid or base.

Distinguishing between weak and strong acids and bases

Practical skills

Tool 1: Experimental techniques — Measuring variables

Inquiry 3: Collecting and processing data — Interpreting results

Now that you know the difference between weak and strong acids and bases, discuss with a partner what physical and chemical properties can be observed in the lab to distinguish between weak and strong acids or bases of the same concentration?

Study skills

When comparing the properties of weak and strong acids or bases, make sure that you are comparing solutions of equal concentration. Concentration and strength are two different variables that together will determine the pH of the solution. Concentration is the ratio of solute:solution, while strength is the degree of ionisation. For example, you can have two solutions — a high concentration of a weak acid and a low concentration of a strong acid — that can have the same pH.

Table 1. A summary of the methods described for strong and weak acids.

	0.1 mol dm ^{−3} HCl(aq)	0.1 mol dm ^{−3} CH ₃ COOH(aq)
[H ⁺ (aq)]	0.1 mol dm ^{−3}	0.0013 mol dm ^{−3}
pH	1.00	2.87
Electrical conductivity	high	low

	0.1 mol dm ⁻³ HCl(aq)	0.1 mol dm ⁻³ CH ₃ COOH(aq)
Relative rate of reaction with magnesium	fast	slow
Relative rate of reaction with calcium carbonate	fast	slow

How can you differentiate between strong and weak bases?

Strong bases have higher pH values and electrical conductivity than weak bases for solutions of the same concentration.

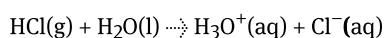
Table 2. Differentiating between strong and weak bases.

	0.1 mol dm ⁻³ NaOH(aq)	0.1 mol dm ⁻³ NH ₃ (aq)
[OH ⁻ (aq)]	0.1 mol dm ⁻³	0.0013 mol dm ⁻³
pH	13–14	10–11
Electrical conductivity	high	low

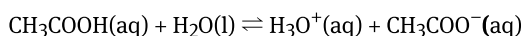
Strength of conjugate acids and bases

We will now look at the strength of the conjugate bases in the reactions of hydrochloric acid (a strong acid) and ethanoic acid (a weak acid) with water.

In the reaction with hydrochloric acid, the water molecule is a much stronger base than the chloride ion. In other words, the water molecule has a much greater tendency to accept a proton than the chloride ion. Therefore, the position of the equilibrium lies to the right and virtually all of the hydrogen chloride molecules will dissociate.



In the reaction of ethanoic acid with water, the ethanoate ion is a much stronger base than the water molecule. The ethanoate ion has a much greater tendency to accept a proton than the water molecule. Therefore, the position of equilibrium lies to the left and relatively few ethanoic acid molecules will dissociate.



In general, strong acids have relatively weak conjugate bases and weak acids have relatively strong conjugate bases.

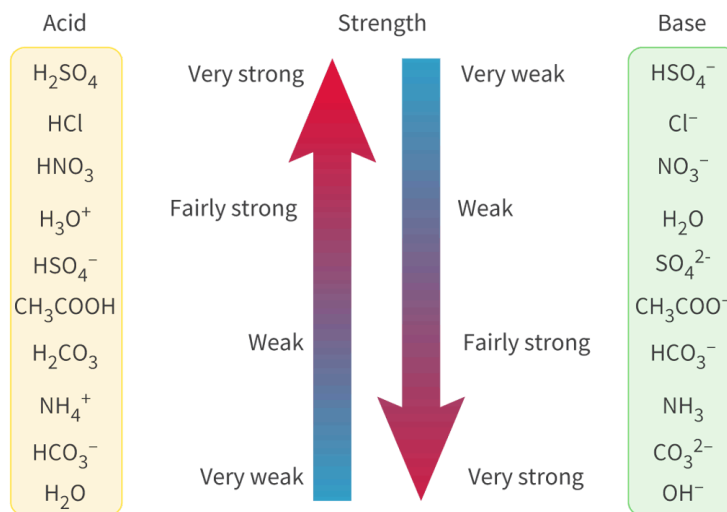


Figure 2. Some common acids and conjugate bases in order of their strengths.

 More information for figure 2

The diagram illustrates a comparison of acids and their conjugate bases with labels indicating their relative strengths. On the left side, a column labeled 'Acid' lists various acids: H_2SO_4 , HCl , HNO_3 , H_3O^+ , HSO_4^- , CH_3COOH , H_2CO_3 , NH_4^+ , HCO_3^- , and H_2O . On the right side, a column labeled 'Base' lists the corresponding conjugate bases: HSO_4^- , Cl^- , NO_3^- , H_2O , SO_4^{2-} , CH_3COO^- , HCO_3^- , NH_3 , CO_3^{2-} , and OH^- .

Between these columns are two large vertical arrows labeled 'Strength,' depicting the relative acid and base strengths. The left arrow indicates acids range from 'Very strong' at the top to 'Very weak' at the bottom. The right arrow indicates conjugate base strengths listed from 'Very weak' at the top to 'Very strong' at the bottom. The arrow direction shows the inverse relationship between acid and conjugate base strength.

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A strong acid will ionise more than a weak acid and the position of equilibrium lies to the right; the conjugate base is relatively weak. Conversely, for a weak acid, the conjugate base is relatively strong, and the equilibrium lies to the left. A similar situation occurs for bases. A strong base has a weak conjugate acid and a weak base has a strong conjugate acid.

Making connections

Knowing that strong acids completely dissociate, while weak acids partially dissociate, how would you expect the equilibrium constants of strong and weak acids to compare? (see [subtopic R2.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

Think about the bonds involved, and based on your knowledge of strong and weak acids, why does the acid strength of the hydrogen halides increase down group 17? (see [subtopic R1.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43468/>)).

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate engagement with issues of global significance
- Demonstrate how to initiate and plan a CAS experience

Have you ever considered the effects that acids can have on the environment?

One such example is acid rain. The combustion of coal in power stations can produce acidic gases such as sulfur dioxide (SO₂). Once in the atmosphere, the SO₂ dissolves in water to produce acid rain. Acid rain can have harmful effects

 ([https://www.eia.gov/energyexplained/coal/coal-and-the-environment.php#:~:text=Several%20principal%20emissions%20result%20from, respiratory%20illnesses%20and?](https://www.eia.gov/energyexplained/coal/coal-and-the-environment.php#:~:text=Several%20principal%20emissions%20result%20from, respiratory%20illnesses%20and?on%20the%20natural%20environment%20and%20can%20also%20damage%20statues,particularly%20those%20made%20of%20limestone%20(calcium%20carbonate).))

A possible CAS experience could be to research the use of coal in your country or region and raise awareness about the effects of its use as an energy source.

Try the activity in the next section to check your understanding of acids and bases.